

MANGALORE REFINERY AND PETROCHEMICALS LIMITED

QUALITY ASSURANCE & MECHANICAL INTEGRITY INSPECTION DOSSIER

Equipment Tag:	P-101A (Booster Pump)	Inspection Report #:	MRPL-NDT-2026-0982
Plant Location:	Unit 01 (HCU) Area 3B	Inspection Date:	2026-09-04
Inspection Type:	Turnaround NDT & Ultrasonic Survey	Lead Inspector:	P. K. Verma (ASNT Level III)
Applicable Code:	API 510 / API 570 / ASME B31.3	Asset Status:	FIT FOR SERVICE (FFS)

1.0 EXECUTIVE SUMMARY & FIT-FOR-SERVICE EVALUATION

Non-Destructive Testing (NDT) comprehensive survey was conducted on primary booster pump P-101A and associated discharge manifold spool 12-CDU-401-HC. Testing encompassed Ultrasonic Thickness Gauging (UT), Liquid Penetrant Examination (PT) of high-stress casing nozzle welds, and Tri-Axial Vibration Spectral Analysis. The equipment satisfies API 570 and API 610 fitness-for-service criteria with calculated remaining corrosion life of 9.4 years.

2.0 ULTRASONIC THICKNESS (UT) GAUGING SURVEY

Ultrasonic thickness measurements performed using Olympus 38DL Plus precision thickness gauge calibrated against ASTM A106 Grade B test step wedge:

Measurement Location Tag	Nominal (mm)	Min Required (mm)	Actual Measured (mm)	Corrosion Rate (mm/yr)	Evaluation Status
P101A-C1: Suction Nozzle Neck	14.20	9.50	13.45	0.08	ACCEPTABLE
P101A-C2: Impeller Volute Casing Top	18.50	12.20	17.80	0.11	ACCEPTABLE
P101A-C3: Discharge Nozzle Flange	16.00	10.80	15.10	0.10	ACCEPTABLE
SPOOL-401-E1: Discharge Elbow 90°	12.70	8.40	11.15	0.16	MONITOR (Erosion)
SPOOL-401-T1: Check Valve Upstream	12.70	8.40	11.85	0.09	ACCEPTABLE
PRV-402-IN: Safety Valve Inlet Nozzle	11.10	7.20	10.60	0.06	ACCEPTABLE

3.0 TRI-AXIAL VIBRATION SPECTRAL ANALYSIS

Vibration telemetry gathered via CSI 2140 Machinery Health Analyzer. Measurements recorded at rated 2,980 RPM under 380 m³/h flow condition:

Bearing Housing Location	1X Peak (40.7 Hz)	2X Harmonic (81.4 Hz)	3X Peak (122.1 Hz)	Overall RMS	ISO 10816-3 Zone
Pump Drive End (DE) Horizontal	1.42 mm/s	0.48 mm/s	0.85 mm/s	2.35 mm/s	Zone A (Good)
Pump Drive End (DE) Vertical	1.10 mm/s	0.35 mm/s	0.62 mm/s	1.82 mm/s	Zone A (Good)
Pump Non-Drive End (NDE) Axial	1.65 mm/s	0.72 mm/s	0.44 mm/s	2.60 mm/s	Zone B (Acceptable)
Motor Drive End (DE) Horizontal	1.25 mm/s	0.30 mm/s	0.20 mm/s	1.65 mm/s	Zone A (Good)

DIAGNOSTIC INTERPRETATION:

The slight elevation in Non-Drive End (NDE) axial vibration (2.60 mm/s) reflects residual hydraulic thrust during throttling operations. FFT spectra indicates zero sub-synchronous oil whirl or bearing cage fault frequencies. Mechanical balance is within API 610 G2.5 balance quality grade.

4.0 MECHANICAL SEAL PLAN 53B PRESSURE BARRIER AUDIT

Seal Parameter Audit	Recorded Value	Acceptable Window	Compliance Verdict
Barrier Fluid Type	Mobil SHC 524 Synthetic PAO	PAO ISO VG 32 Synthetic	VERIFIED
Accumulator Bladder Nitrogen Precharge	18.2 bar gauge at 25°C	18.0 ± 0.5 bar gauge	VERIFIED
Barrier Loop Working Pressure	22.5 bar gauge	20.0 to 24.0 bar gauge	VERIFIED
Differential over Stuffing Box	2.1 bar positive delta	> 1.5 bar positive delta	SAFE (No Reverse Flow)
24-Hour Barrier Fluid Consumption	45 ml / 24 hours	< 120 ml / 24 hours	EXCELLENT
Barrier Fluid Temperature	48.5 °C at cooler outlet	< 65.0 °C	NORMAL

5.0 INSPECTION PUNCH LIST & CORRECTIVE WORK ORDERS

Item #	Asset Component	Defect / Condition Observed	Target Work Order	Resolution Deadline
PNC-01	Discharge Elbow SPOOL-401-E1	Localized wall erosion from particulate turbulence. Re-gauge in 90 days.	WO-889021 (PM02)	2026-12-01
PNC-02	Lube Oil Sight Glass	Minor discoloration on outer viewing acrylic. Replace with Pyrex glass.	WO-889045 (PM01)	Next Routine PM
PNC-03	Earth Grounding Bond	Grounding strap copper braid showing surface oxidation (< 0.2 ohms).	WO-889088 (PM01)	2026-09-30

6.0 FINAL CERTIFICATION & ENGINEERING ENDORSEMENT

Based upon non-destructive examination, ultrasonic thickness survey, and dynamic vibration baseline analysis, centrifugal pump **P-101A** is certified **FIT FOR PURPOSE** to operate up to design pressure of 500 PSI and maximum flow capacity of 380 m³/h in accordance with API 510 and OISD standards.

Inspected By: P. K. Verma, Lead NDT Inspector ASNT Level III (UT/PT/RT/VT) Date: 2026-09-04	Reviewed & Verified: M. Ananthakrishnan Senior Reliability Engineer Date: 2026-09-05	Approved For Service: Dr. S. K. Roy Chief General Manager (Refining Maintenance) Date: 2026-09-06
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